

$$3.2] \langle f, g \rangle = \int_1^5 f(x)g(x)dx$$

$$a) V_0 = (1, x, x^2) \quad \text{I) } u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{\int_1^5 1 dx}} = \frac{1}{2}$$

$$\text{II) } v_2 - w_2 = x - \langle \frac{1}{2}, x \rangle \cdot \frac{1}{2} = x - 3$$

$$L \int_1^5 \frac{1}{2} x = 4.6$$

$$u_2 = \frac{x-3}{\|x-3\|} = \frac{\sqrt{3}}{4} (x-3)$$

$$L \int_1^5 (x-3)^2 dx$$

$$\text{III) } v_3 - w_3 = x^2 - \langle \frac{1}{2}, x^2 \rangle \frac{1}{2} - \langle \frac{\sqrt{3}}{4} (x-3), x^2 \rangle \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$L \int_1^5 \frac{1}{2} x^2 dx \quad L \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 dx$$

$$= x^2 - 6x + \frac{23}{3}$$

$$u_3 = \frac{v_3 - w_3}{\|v_3 - w_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{216}{45}}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{24}{5}}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{24}{5}}}$$

$$= \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{24}{5}}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{24}{5}}}$$

$$B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{x^2-6x+\frac{23}{3}}{\sqrt{\frac{286}{45}}} \right)$$

$$b) [W_B] = \begin{pmatrix} \langle u_1, v \rangle \\ \langle u_2, v \rangle \\ \langle u_3, v \rangle \end{pmatrix} \quad v = \sin x$$

$$= \left(\int_1^5 \frac{1}{2} \sin x \, dx, \int_1^5 \frac{\sqrt{3}}{4} \sin x (x-3) \, dx, \int_1^5 \sin x \frac{x^2-6x+\frac{23}{3}}{\sqrt{\frac{286}{45}}} \, dx \right)^T$$

$$\approx 0,128 \quad \approx -1,5 \quad \approx -0,125$$

$$\sin x \approx 0,128 \cdot 0,5 + \frac{\sqrt{3}}{4} \cdot (-1,5) \cdot (x-3) - 0,125 \cdot \frac{(x^2-6x+\frac{23}{3})}{\sqrt{\frac{286}{45}}}$$

$$= -0,05x^2 - 0,335x + 1,61$$

$$c) \|\sin x - (-0,05x^2 - 0,335x + 1,61)\| = \sqrt{\int_1^3 (\sin x - (-0,05x^2 - 0,335x + 1,61))^2 \, dx} = 0,32$$